Student Management System (Flutter with Hive)

Student Details

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Course: B.Tech IT
Subject: Flutter Development

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1. Introduction

The Student Management System is a simple Flutter application developed to demonstrate local data storage in mobile applications. The application allows users to store, retrieve, and update student information using Hive local database.

This project is created as part of an academic activity to understand how Flutter applications can manage data locally without using external servers or cloud databases.

The system provides basic functionality for managing student records such as adding new students, viewing saved students, and updating student information.

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2. Objectives

• To understand Flutter application development
• To implement local data storage using Hive database
• To demonstrate basic CRUD operations (Create, Read, Update)
• To manage student information within a mobile application

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3. Technologies Used

Flutter – for building the mobile application UI and logic
Dart – programming language used by Flutter
Hive – lightweight local database for storing data
GitHub – for version control and project hosting

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4. System Features

Add Student
Users can enter student details such as name, enrollment number, and course, and store them in the local database.

View Student List
The application displays all saved students in a list format.

Update Student
Users can edit and update student information stored in the database.

Local Data Storage
All data is stored locally using the Hive database and remains saved even after the application is closed.

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5. System Workflow

Step 1
The user opens the Flutter application.

Step 2
The user selects the option to add a new student.

Step 3
The user enters student details such as name, enrollment number, and course.

Step 4
The data is stored locally in the Hive database.

Step 5
The user can view the list of saved students.

Step 6
If required, the user can update the student details.

Step 7
The updated data is saved again in the local database.

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6. Application Architecture

The application follows a simple structure consisting of:

Main Application File (main.dart)
Initializes the Hive database and launches the application.

Model
Defines the structure of the student data.

Screens
Home Screen – main navigation page
Add Student Screen – for entering student data
Student List Screen – displays stored students

Database
Hive database is used for storing student records locally.

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7. GitHub Repository

The complete source code of the project is uploaded on GitHub.

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8. Conclusion

The Student Management System demonstrates how Flutter applications can store and manage data locally using Hive. The project successfully implements basic CRUD operations and provides a simple interface for managing student records.

This project helps in understanding the integration of Flutter with local databases and the use of GitHub for project version control and collaboration.

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9. Screenshots

Student Management

Add Student

View Students

Add Student

Student Name
tanmay

Enrollment Number
20230905050507

Course
CSE

Save

Student List

tanmay
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